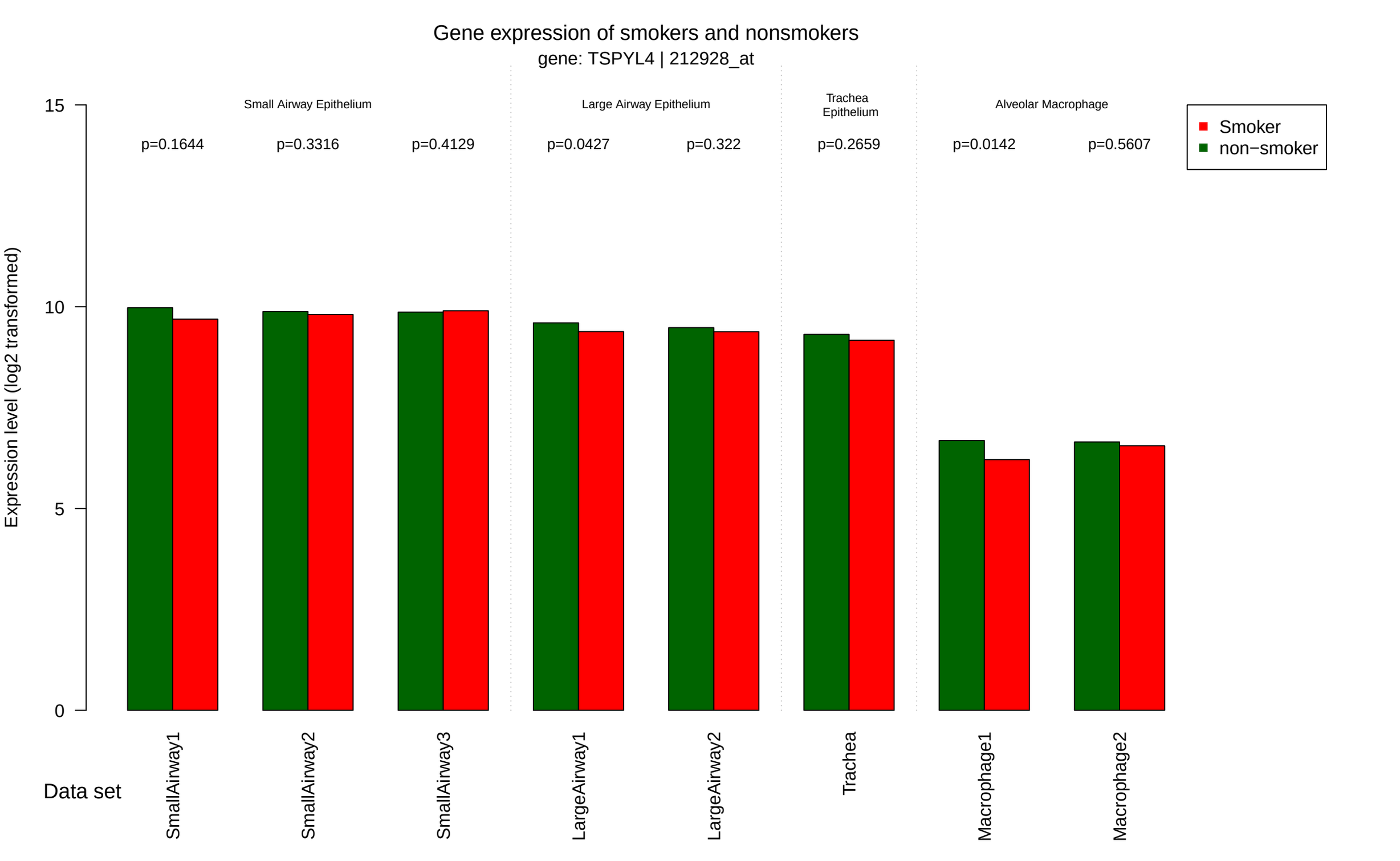
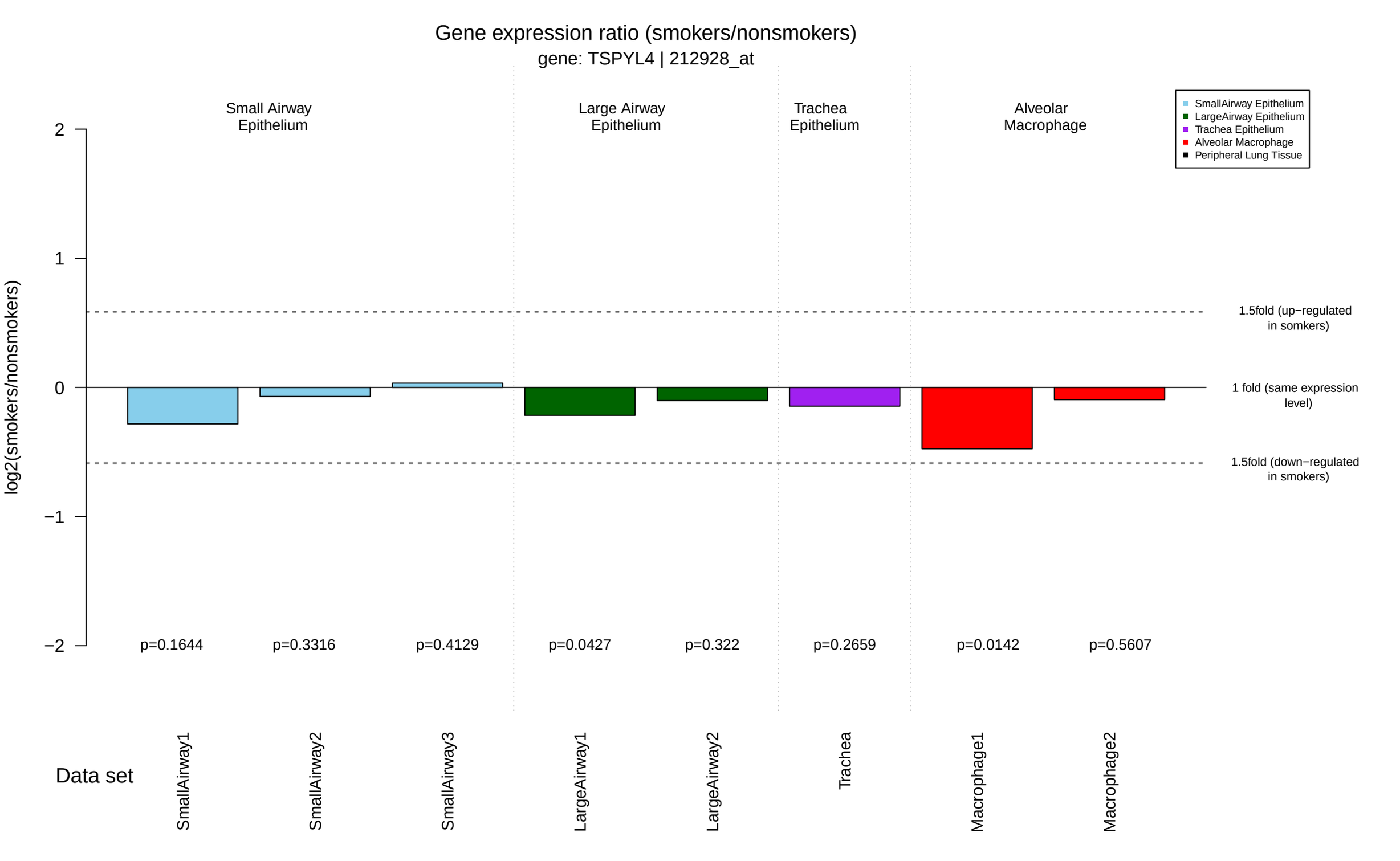
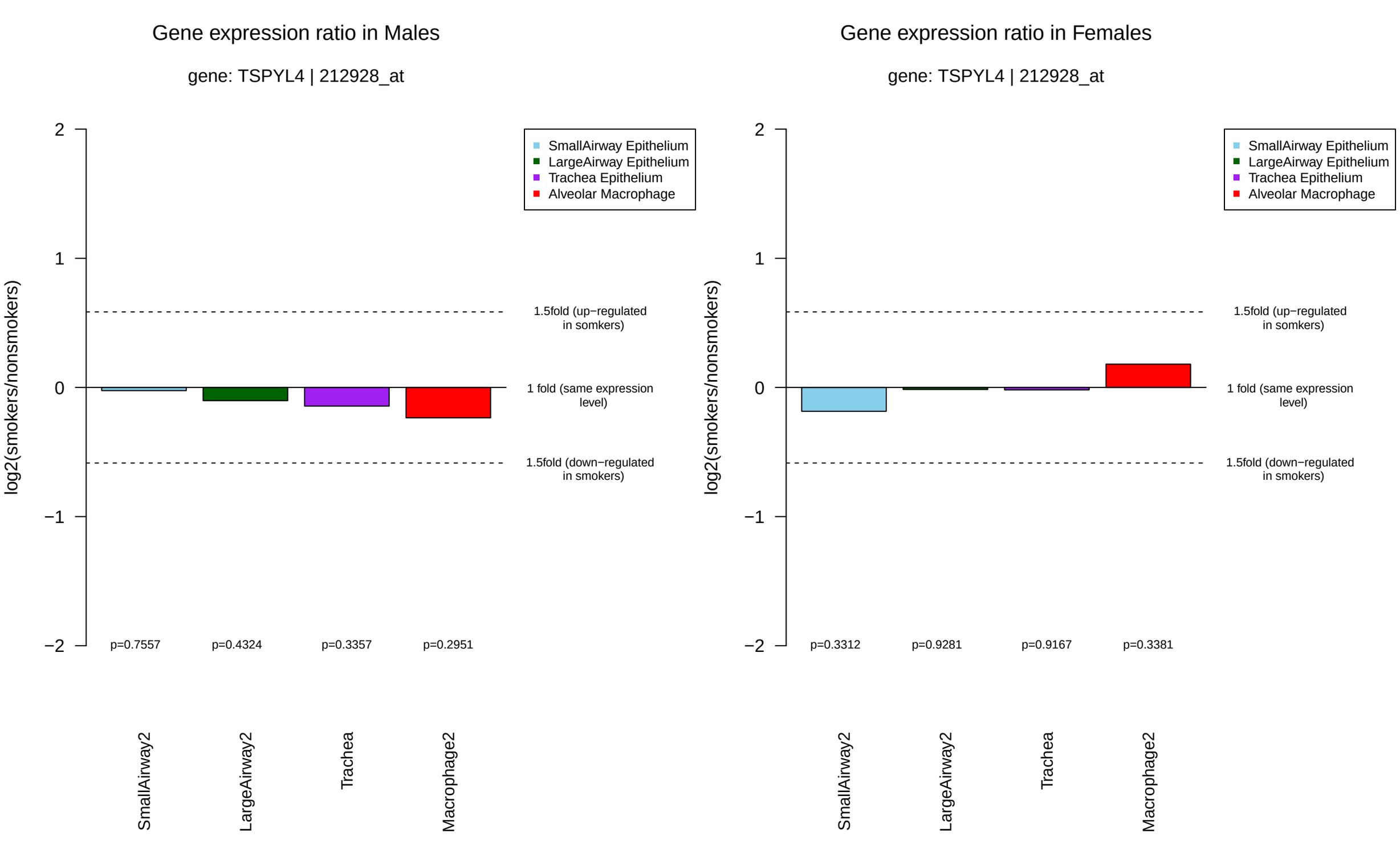
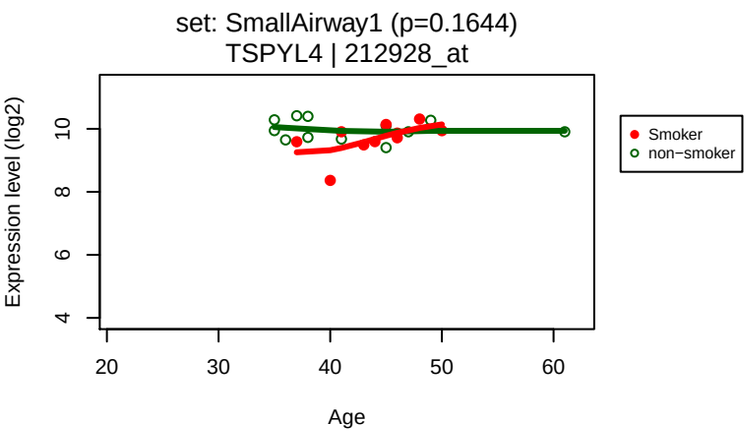
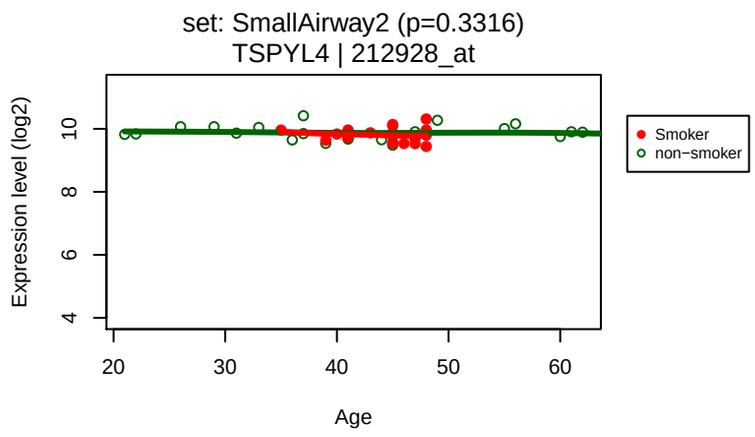
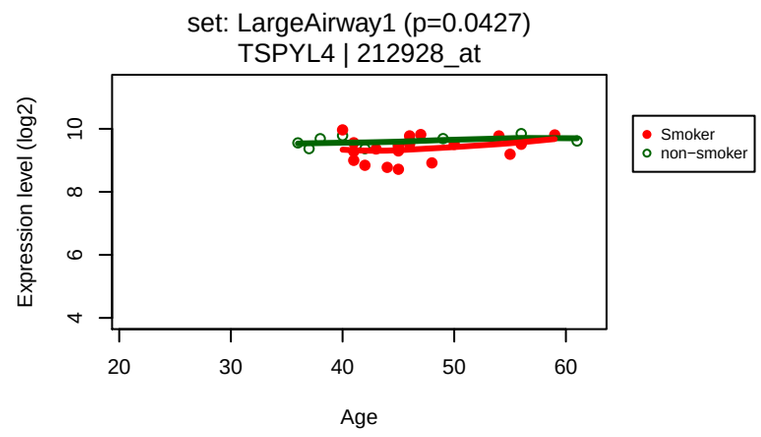

Gene expression of smokers and nonsmokers
gene: TSPYL4 | 212928_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1644
p=0.3316
p=0.4129
p=0.0427
p=0.322
p=0.2659
p=0.0142
p=0.5607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

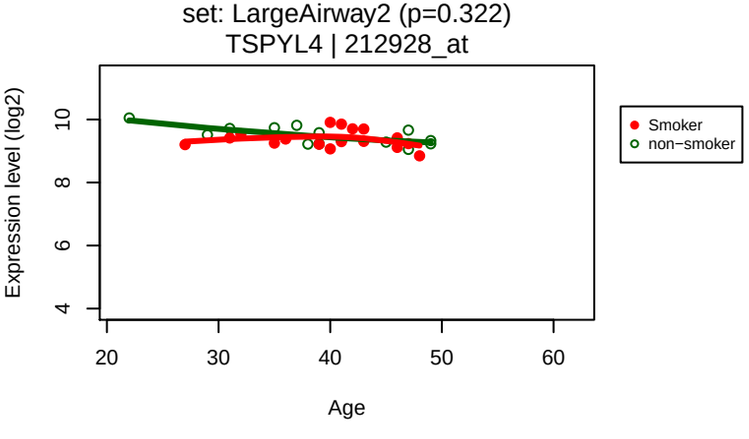

Gene expression ratio (smokers/nonsmokers)
gene: TSPYL4 | 212928_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1644
p=0.3316
p=0.4129
p=0.0427
p=0.322
p=0.2659
p=0.0142
p=0.5607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

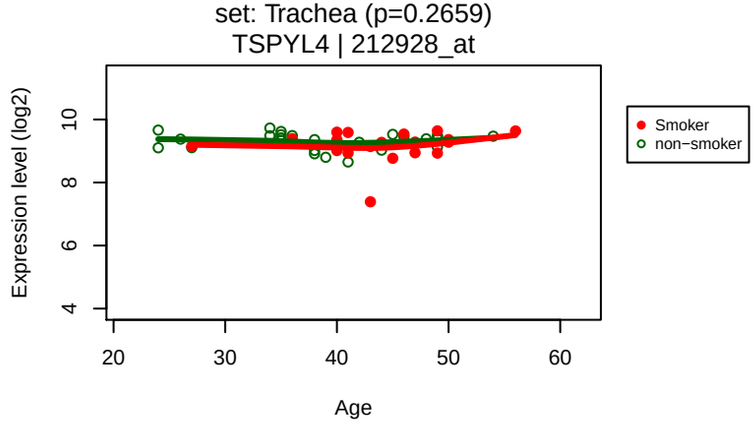

Gene expression ratio in Males
gene: TSPYL4 | 212928_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7557
p=0.4324
p=0.3357
p=0.2951
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPYL4 | 212928_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3312
p=0.9281
p=0.9167
p=0.3381
SmallAirway2
LargeAirway2
Trachea
Macrophage2

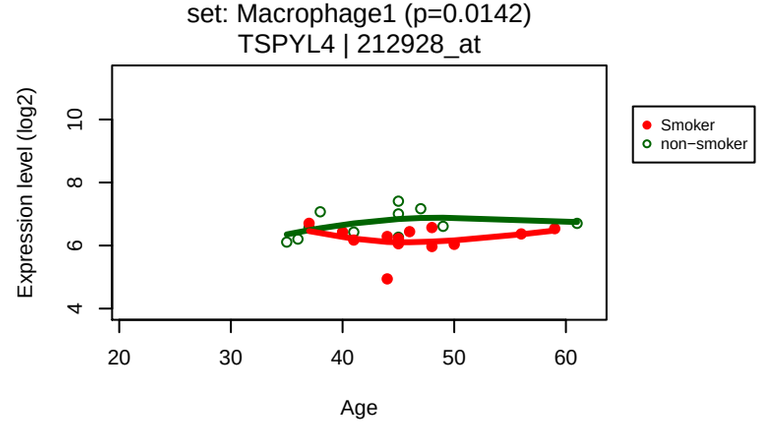

set: SmallAirway1 (p=0.1644)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker

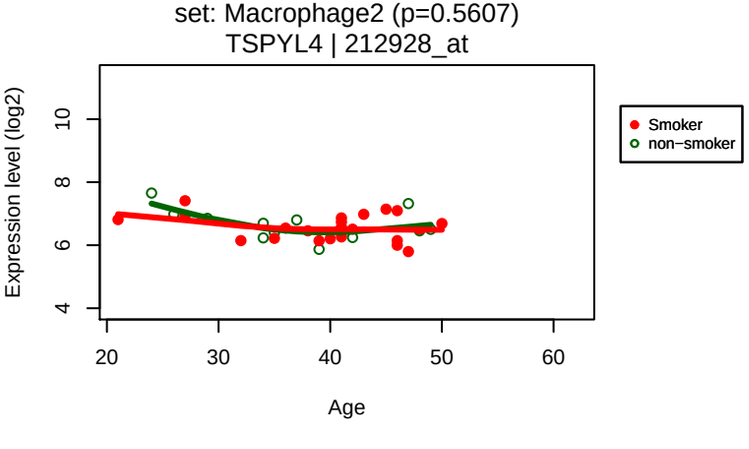

set: SmallAirway2 (p=0.3316)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker

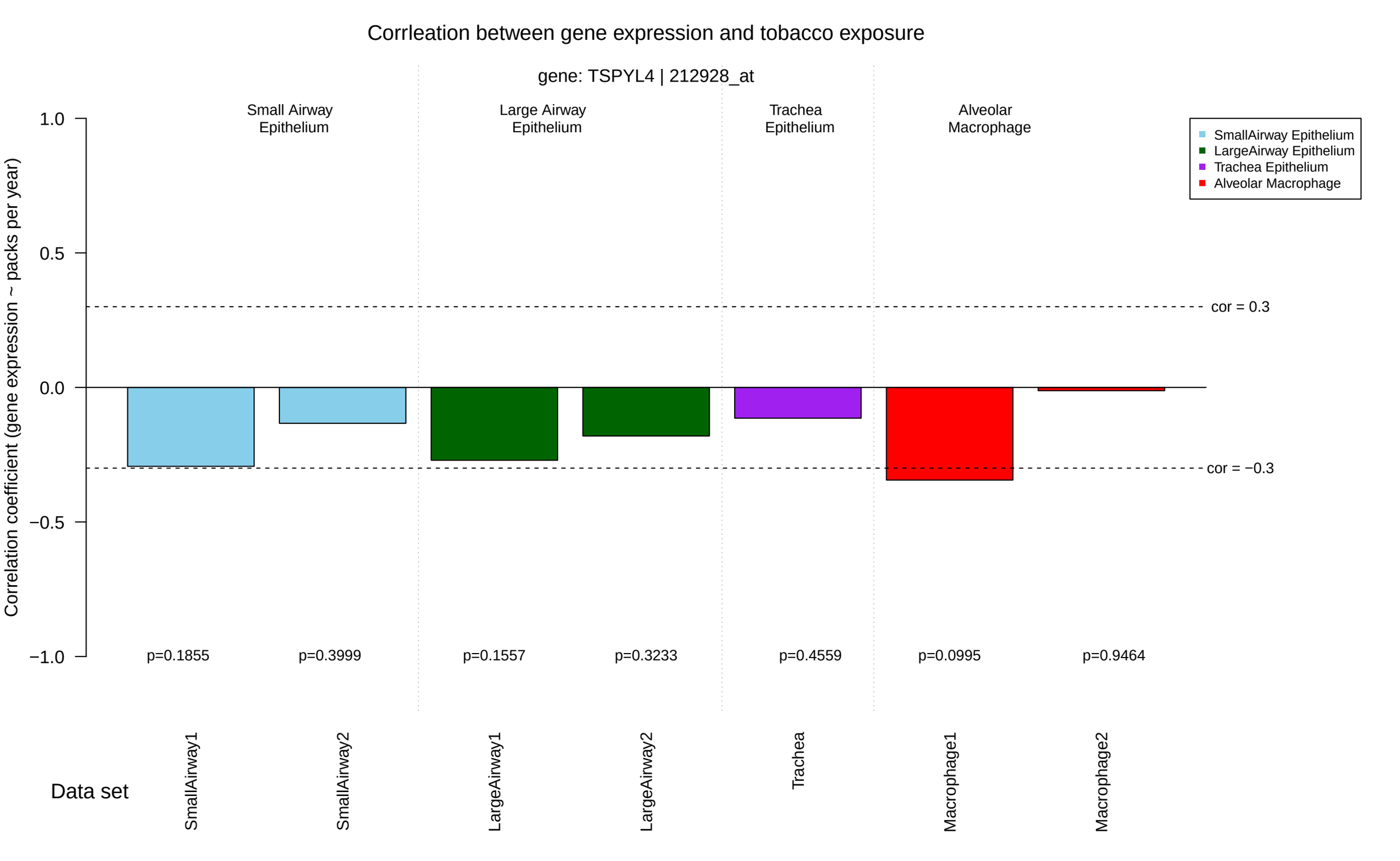

set: LargeAirway1 (p=0.0427)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.322)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2659)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0142)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5607)
TSPYL4 | 212928_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPYL4 | 212928_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1855
p=0.3999
p=0.1557
p=0.3233
p=0.4559
p=0.0995
p=0.9464
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2